# A Multimodal Cross-Domain Segmentation Network of Remote Sensing Imagery With Multilevel Deep Cross-Fusion and Adversarial Domain Adaptation

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***Abstract*— Deep learning techniques have recently achieved significant success in semantic segmentation for Earth observation tasks using single-modality data in a specific region. However, such optimal conditions are seldom found in real-world applications, leading to subpar performance and limited generalization of existing models when faced with multimodal cross-domain scenarios. To tackle this issue, we introduce a novel segmentation network for remote sensing imagery (RSI) called MC-Seg, which incorporates multilevel deep cross-fusion and adversarial domain adaptation (DA). The MC-Seg framework begins by performing multilevel deep cross-fusion of multimodal data through a complete feature extraction (CFE) module and three fusion levels: feature, modal, and layer. This approach ensures the thorough utilization of features from various modalities. Following this, the framework integrates a local–global adversarial DA (LGADA) module to minimize the domain gap, effectively aligning the source and target domains despite the differences in RSI data representations. Experimental validation on the C2Seg dataset shows that our proposed method outperforms existing state-of-the-art techniques.**

***Index Terms*— Cross-domain segmentation, deep cross-fusion, domain adaptation (DA), multimodal.**

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## I. Introduction

SEMANTIC segmentation of remote sensing imagery (RSI) is a critical and challenging task in the realm of RSI interpretation. Recent advancements in satellite remote sensing (RS) technology have broadened its applications across various fields, including geological observation [1], disaster prevention and mitigation [2], and urban planning [3]. The increasing number of diverse RS satellites capable of capturing high-resolution images has significantly expanded the sources for RSI collection and dataset scales, thereby enriching the availability of RSI data. In practical applications, RSIs from different sensors often exhibit multimodal characteristics, encompassing optical, infrared, and radar images. This multimodal data captures various aspects of the Earth's surface, such as land-cover types, vegetation coverage, and land use. Utilizing these multimodal datasets for image segmentation can yield more comprehensive and accurate insights, which can help address more practical scenarios.

In the early stages, image segmentation techniques utilizing multimodal data primarily relied on manual feature extraction methods [4]. Traditional approaches, such as component replacement, geostatistical analysis, and sparse coding, often resulted in detail loss and a significant decline in segmentation accuracy [5]. This decline is mainly due to the complexity of the feature extraction algorithms and the fusion rules employed [6]. Furthermore, variations in brightness and contrast caused by data acquisition from different sensors further complicate the analysis. Consequently, conventional processing and analysis methods, particularly those based on expert systems, face challenges in effectively handling large-scale, multimodal, and high-resolution RSI data. Therefore, achieving accurate and efficient RSI segmentation using multimodal data remains a significant challenge.

The advent of deep learning technology offers a promising solution to this issue. Advanced models, such as convolutional neural networks (CNNs), have been widely adopted in RS applications, including land-use and land-cover mapping [7], [8]. For instance, Fu et al. [9] proposed a block-based object-oriented image segmentation method that leverages CNNs to automatically extract complex features from raw image data, facilitating end-to-end classification of segmented objects and mitigating the uncertainties linked to manual feature extraction. These models have proved effective in practical

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applications, especially within consistent regional contexts where the types, characteristics, and spatial distributions of surface elements align closely. However, most existing models are designed for single-modality data in the same regional context. Their handling of multimodal data is often restricted to early fusion [10], where diverse modalities are integrated into a single modality at an initial stage for further processing. This manner fails to fully exploit the potential of complementary information in multimodal data. Some research efforts [11], [12] have sought to implement fusion operations during the middle and later stages of the model to better utilize this complementary information. For instance, Liu et al. [13] proposed a mid-fusion segmentation network for multimodal RS images, termed SSAFNet, which processes optical and synthetic aperture radar (SAR) images through two independent encoder–decoder streams in parallel to extract high-level semantic features from different modalities. These features are then integrated into the network for deep fusion, resulting in joint image segmentation. Sun et al. [14] proposed a segmentation network for natural color, infrared images, and digital surface models using a modified DeepUNet, successfully merging shared and unique features through a series of segmentation and fusion processes. Nevertheless, most of these models heavily rely on annotated image datasets. As RS technology advances, the scale of datasets has grown significantly, making it impractical to create annotated datasets for every distinct region. Consequently, these models face limitations in addressing the complex challenges posed by multimodal data and cross-domain environments.

The deep fusion of multimodal RS data enhances the expressiveness of various modalities, improves segmentation accuracy, and mitigates the effects of domain gaps in cross-domain tasks [15], [16]. Different domain adaptation (DA) techniques can effectively bridge the gap between the source and target domains. By merging these two strategies, it is possible to develop more sophisticated models suitable for mixed-application scenarios. This combination facilitates the comprehensive use of multimodal data, mitigates semantic differences across domains, and improves the generalization capabilities for cross-domain tasks. For instance, Tasar et al. [17] proposed a novel approach called DAugNet for unsupervised, multisource, multitarget, and DA in satellite imagery. This method comprises a classifier and a data augmenter, where the data augmentation component is a shallow network that performs style transfer among multiple satellite images in an unsupervised manner. This provides varied data to the classifier, enhancing its resilience to significant differences in data distribution across domains.

Despite the advancements made in multimodal cross-domain RSI semantic segmentation, existing research still faces several limitations, including the following.

1) *Lack of Close Integration Between Feature Extraction and Fusion:* In many current models, feature extraction and fusion are treated as separate, sequential processes. This often results in missed opportunities to leverage information from features at different depths, leading to inadequate cross-modal feature extraction and less effective fusion outcomes.
2) *Neglect of Modality-Specific Heterogeneity:* Different imaging mechanisms result in varying representations of the same scene across modalities. Utilizing a common backbone for feature extraction from each modality limits the full exploitation of complementary information, resulting in incomplete feature data before DA operations.
3) *Inadequate Alignment of Features Between the Source and Target Domains:* A one-stage global alignment method, such as adversarial learning, can lead to negative information transfer, which may diminish the performance of well-aligned regions during the early training phases.

To address the aforementioned issues, we propose MC-Seg for the multimodal cross-domain RSI segmentation that leverages multilevel deep cross-fusion and adversarial DA. The framework operates in two main stages. First, MC-Seg performs multilevel deep cross-fusion of multimodal data through several modules, including the complete feature extraction (CFE) module, the feature-level fusion (FLF) module, the modal-level fusion (MLF) module, and the layer-level fusion (LLF) module. This ensures the comprehensive utilization of features from different modalities. Second, it incorporates a local–global adversarial DA (LGADA) module to minimize the domain gap, effectively bridging the discrepancies between the source and target domains caused by variations in RS imagery representations.

The main contributions of this study are as follows.

1) To closely combine feature extraction and fusion and increase the feature representation capability of the network, we devise the CFE module to fully extract the local–global features from single-modality data and the FLF module to fuse the features from other modalities in 3-D.
2) The MLF module is designed to thoroughly investigate the complementary features between different modalities. It calculates the cosine distance between multimodal data and applies this to the original data to establish heterogeneous correspondence relationships, thereby increasing confidence in features identified as complementary.
3) The LGADA module is presented to ensure complete alignment of features between the source- and target-domain images, effectively reducing the gap between them. The DA for the target domain is achieved by utilizing local alignment to inform global alignment.
4) The proposed MC-Seg is evaluated using two benchmark multimodal cross-domain RS datasets, C2Seg-AB and C2Seg-BW. The experimental results consistently demonstrate the superiority of the method.

The remainder of the article is structured as follows. Section II reviews the related work. Section III presents our proposed network in detail. Experiments were conducted on various datasets and extensively discussed and analyzed in Section IV. Finally, Section V summarizes the study.

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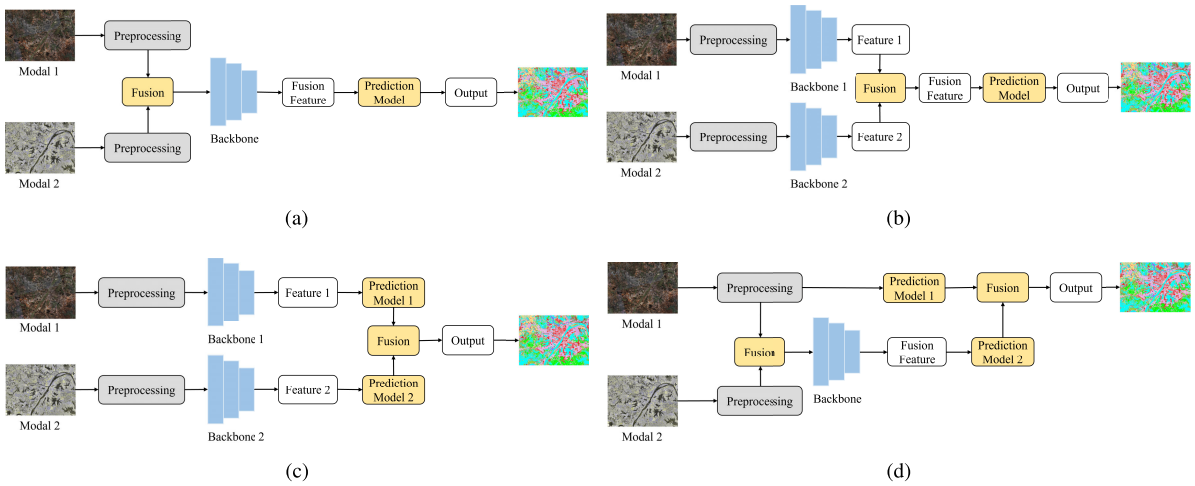



Fig. 1. Four typical multimodal fusion paradigms. (a) Early fusion scheme, (b) mid-fusion scheme, (c) late fusion scheme, and (d) hybrid fusion scheme.

## II. Related Work

RSI semantic segmentation typically involves managing large datasets, a task that traditional processing techniques often struggle to handle effectively. Deep learning is currently the most advanced and widely adopted technology for addressing this challenge. Based on various research contexts, we categorize these methods into three groups: multimodal same-domain, single-modality cross-domain, and multimodal cross-domain, which will be discussed in detail.

### A. Multimodal Same-Domain Scenarios

As the volume of relevant RS data continues to grow, the diversity of data modalities has also increased. Different modalities can be integrated and analyzed together through data fusion (DF) techniques to create more comprehensive scene descriptions [18]. In the realm of RSI semantic segmentation, multimodal fusion methods have become a crucial approach. These strategies can be divided into four types: early fusion [Fig. 1(a)], mid-fusion [Fig. 1(b)], late fusion [Fig. 1(v)], and hybrid fusion [Fig. 1(d)] [19]. The primary difference among these methods lies in the stage at which the modalities are combined within the model.

By leveraging complementary information from various modalities, multimodal semantic segmentation demonstrates improved performance compared to single-modality approaches [20]. However, traditional CNN-based multimodal fusion techniques encounter difficulties in modeling long-range relationships between multiscale feature maps from different modalities in RSI data. To address this issue, Ma et al. [21] proposed a multiscale cross-modal fusion network utilizing a Transformer architecture, which enhances the modeling of local–global contextual information. Additionally, to tackle the limitations of previous networks in effectively utilizing the diverse and rich information present in each modality, as well as addressing intermodal heterogeneity and complementarity, Xiao et al. [22] introduced an adaptive multimodal feature fusion module aimed at achieving a comprehensive fusion of features from different modalities. Despite these advancements, the effectiveness of these methods is often restricted to the specific test set of the trained dataset and does not generalize well to other datasets with similar characteristics.

### B. Single-Modality Cross-Domain Scenarios

Applying a segmentation model that has been trained on a dataset (the source domain) to a different dataset (the target domain) often leads to a decline in performance due to the domain gap. The DA techniques have been shown to effectively reduce the semantic discrepancies between the source and target domains [23], [24]. Benjdira et al. [25] designed an algorithm utilizing generative adversarial networks (GANs) to alleviate the effects of variations in feature distribution. Building on this foundation, Chen et al. [26] proposed an unsupervised DA (UDA) framework that employs adversarial learning for the semantic segmentation of high-resolution RSIs. Following this, Zhu et al. [27] created an adversarial learning strategy for the output space aimed at minimizing the domain distribution differences between the source and target domains, thereby mitigating the effects of domain shifts. However, most existing DA methods are unidirectional, which limits their effectiveness as they do not account for the reverse process and fail to fully leverage information from other domains. To overcome these limitations, Cai et al. [28] proposed a novel unsupervised bidirectional DA framework. This framework learns to transform source images into pseudo-target images while adapting the classifier to the target domain in the source-to-target direction. Conversely, it converts target images into pseudo-source images and refines the source classifier in the opposite direction.

### C. Multimodal Cross-Domain Scenarios

The integration of multimodal information has been demonstrated to enhance segmentation accuracy [15], [16]. Furthermore, various DA techniques can remarkably diminish the disparities between the source and target domains. By merging these two approaches, the performance of multimodal cross-domain semantic segmentation tasks can be greatly improved [29], [30]. For example, Hu et al. [31] introduced a new multimodal UDA method called MMADT,

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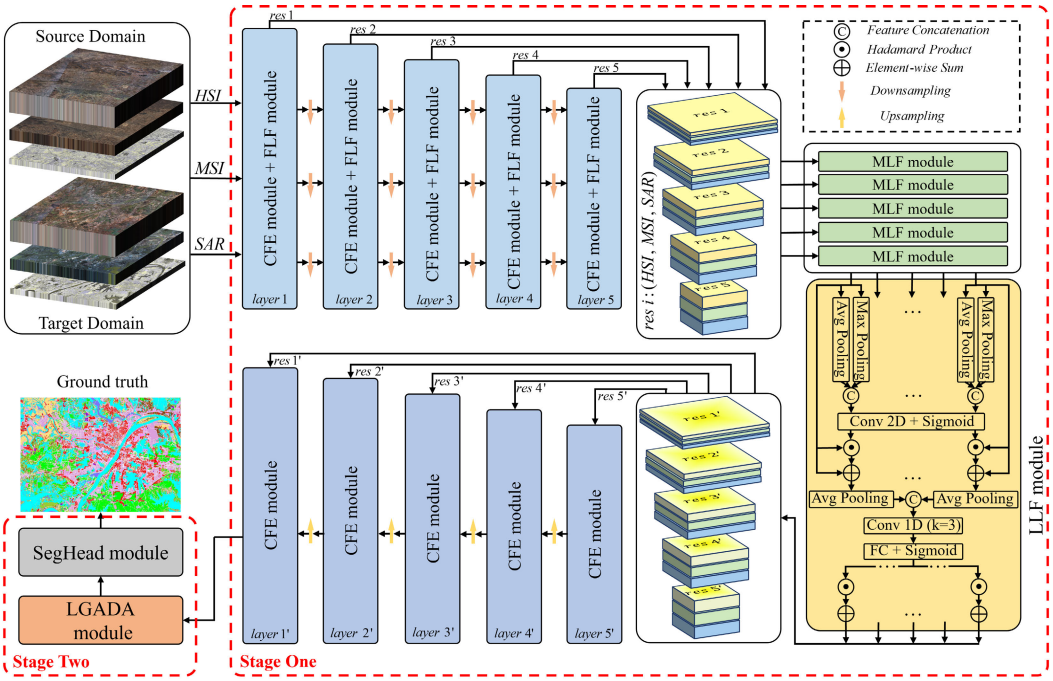



Fig. 2. Flowchart of the proposed MC-Seg, which comprises the CFE module, FLF module, MLF module, LLF module, LGADA module, and SegHead module.

which includes a deep fusion block to recalibrate information and utilizes depth adversarial training to address the disparities between the source and target domains. To enhance the generalization of models across diverse regions, Hong et al. [32] proposed a high-resolution multimodal DA model. This model effectively preserves the spatial topological structure of scenes from the source domain by employing a parallel fusion of high- and low-resolution features, while also bridging the gap caused by significant differences in RSI representations between the source and target domains through adversarial learning.

Recent studies show that scholars have achieved promising results in cross-domain segmentation using multimodal RS data. However, the lack of sufficient integration between semantic segmentation and DA technologies has resulted in performance limitations in multimodal cross-domain semantic segmentation tasks. Consequently, Section III will concentrate on the design and integration of multimodal deep fusion networks with adversarial DA networks.

## III. Methodology

### A. Overall Architecture

As illustrated in Fig. 2, the proposed MC-Seg network consists of two stages. The first stage involves multimodal data feature extraction and fusion through an encoder–decoder structure, incorporating the CFE and three-level fusion modules: feature-level, modal-level, and layer-level. The second stage is dedicated to DA, which comprises an LGADA module and a segmentation head (SegHead) module. This stage employs adversarial learning techniques to align the features of the source and target domains, while also addressing the rebalancing and adjustment of the target domain's feature representation to ensure compatibility with the source-domain features. Furthermore, the loss function from this second stage can be backpropagated to modify the weight parameters of the first stage, creating a cohesive training loop.

### B. Stage One: Multimodal Feature Extraction and Fusion Network Based on Encoder–Decoder

In the first stage, the network employs the CFE, FLF, MLF, and LLF modules, along with continuous downsampling, to extract and integrate features from various modalities across multiple scales. The resulting fused multiscale feature maps are then processed by the decoder, which is based on the CFE module, for upsampling through skip connections.

*1) CFE Module:* In RSI segmentation tasks, the variability in size and shape of the segmented objects poses significant challenges. To leverage the benefits of both CNN and Transformer architectures, we developed the CFE module. This module consists of three branches: a local feature extraction branch utilizing CNNs, a global feature extraction branch that employs an enhanced Transformer, and a residual connection to maintain feature integrity. This design allows for thorough feature extraction while keeping computational demands low [33]. The lower half of Fig. 3 depicts the structure of the CFE module.

For the local feature extraction branch, a sequence of lightweight $3\times3$ convolution blocks is implemented to decompose larger receptive fields. The branch incorporates three consecutive dilated convolution blocks with dilation rates of 1, 2, and 3, each using a kernel size of 3. The number of filters in these convolutional layers increases progressively, following the order $((C_{\text{out}}/6), (C_{\text{out}}/3), (C_{\text{out}}/2))$. The outputs from these three convolution blocks are then concatenated to create spatial features with varying receptive fields, and a residual connection is included to mitigate feature degradation.

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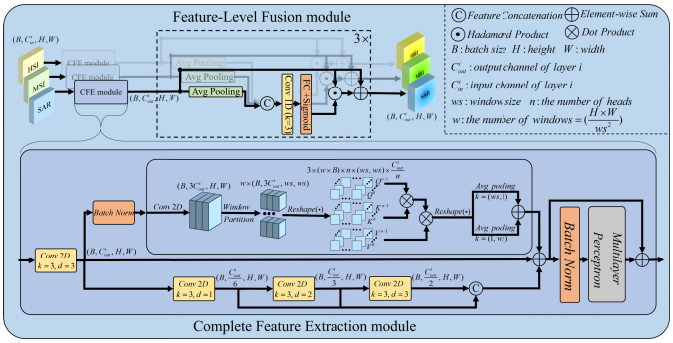



Fig. 3. Illustration of the proposed CFE module and FLF module. The multimodal data is first fully extracted information from the internal features of each modality by the parallel CFE module, then fused with other modal data features by the FLF module, and sent to the next layer.

In the global feature extraction branch, the input dimensions are first adjusted to $(B, 3C_{\text{out}}, H, W)$ using a $1 \times 1$ convolutional layer. A window partitioning strategy is then applied, resulting in a set of feature maps represented as $\mathbb{R}^{((B\times w),3C_{\text{out}},ws,ws)}$, where $ws$ is empirically set to 8 as the window size. These feature maps are subsequently rearranged to form $\mathbb{R}^{(3\times(B\times w)\times n\times(ws\times ws)\times(C_{\text{out}}/n))}$ for self-attention computation. This allows for the flow of information within the window. Following this, a crossed average pooling layer applied both horizontally and vertically is utilized to facilitate effective interaction between windows. By expanding this approach to $w$ windows, the model can capture remote dependencies between any two windows, thereby enhancing the ability to grasp the global context.

*2) FLF Module:* All modalities share a uniformly structured CFE module with independent weights. At this point, the network focuses solely on learning the feature distribution within each modality. To avoid issues associated with directly concatenating data from different modalities, which can create significant gaps, we designed the FLF module. As shown in the upper half of Fig. 3, this module enhances the extraction of single-modality data by fusing the features of other modalities across the row, column, and channel dimensions. The calculation process can be expressed with the following equations:

$$\mathbf{X}_C = \text{Cov1D}(\text{Concat}(\text{AvgP}_C(\mathbf{X}^0, \mathbf{X}^1, \mathbf{X}^2)), \tag{1}$$

$$\mathbf{X}^{m'} = \mathbf{X}^m + \mathbf{X}^m \odot \sigma(FC_C^m(\mathbf{X}_{\text{C}})), m \in (0, 1, 2), \tag{2}$$

$$\mathbf{X}_H = \text{Conv1D}(\text{Concat}(\text{AvgP}_H(\mathbf{X}^{0'}, \mathbf{X}^{1'}, \mathbf{X}^{2'}), \tag{3}$$

$$\mathbf{X}^{m''} = \mathbf{X}^{m'} + \mathbf{X}^{m'} \odot \sigma(FC_H^m(\mathbf{X}_{\text{H}})), m \in (0, 1, 2), \tag{4}$$

$$\mathbf{X}_W = \text{Conv1D}(\text{Concat}(\text{AvgP}_W(\mathbf{X}^{0''}, \mathbf{X}^{1''}, \mathbf{X}^{2''})), \tag{5}$$

$$\mathbf{X}^{m'''} = \mathbf{X}^{m''} + \mathbf{X}^{m''} \odot \sigma(FC_W^m(\mathbf{X}_{\text{W}})), m \in (0, 1, 2) \tag{6}$$

where $\mathbf{X}^{\text{m}}$ denotes the data for the $m$th modality. $\text{AvgP}_C$, $\text{AvgP}_H$, and $\text{AvgP}_W$ represent average pooling layers that retain only the dimensions of the channel, column, and row, respectively. The term Concat signifies a concatenation operation, while Conv1D indicates the 1-D convolution. The functions $\text{FC}_C^m$, $\text{FC}_H^m$, and $\text{FC}_W^m$ represent FC layers with output dimensions equal to the dimensions of the channel, column, and row of the data of the $m$th modality, respectively. $\sigma$ denotes the Sigmoid function, and $\odot$ denotes element-wise multiplication. This approach enables multimodal data to learn different feature information across 3-D. By maintaining the features of each modality while incorporating features from others, the CFE module in the subsequent layer can extract new features more comprehensively.

*3) MLF Module:* Processing multimodal data presents a significant challenge compared to single-modality data, particularly in identifying and fully utilizing complementary features across different modalities. To address this, we propose an MLF module inspired by the Transformer architecture, which seeks to uncover complementary features among various modality data.

As depicted in Fig. 4, let $\mathbf{X} = [x^0, x^1, x^2] \in \mathbb{R}^{B\times 3C\times H\times W}$ represent the multimodal high-order features received by the MLF. Initially, a $3 \times 3$ convolution is applied to the data of each modality for linear transformation, yielding $x^{0'}, x^{1'}, x^{2'}$. Next, a group convolution with $g = 3$ is performed on the original input $\mathbf{X}$, resulting in $\mathbf{X}'' = [x^{0''}, x^{1''}, x^{2''}]$. The traditional window-divide method is then applied, uniformly splitting each feature map of size $H \times W$ into $w$ windows of size $ws \times ws$, producing intermediate values $X^{0'}, X^{1'}, X^{2'} \in \mathbb{R}^{[w\times(B,C,ws,ws)]}$ and $[X^{0''}, X^{1''}, X^{2''}] \in \mathbb{R}^{[w\times(B,3C,ws,ws)]}$. The window size $ws$ is also empirically set to 8. These features are then rearranged into $\overline{X}^{0'}, \overline{X}^{1'}, \overline{X}^{2'} \in \mathbb{R}^{[(w\times B)\times n\times(ws,ws)\times(C/n)]}$ and $[\overline{X}^{0''}, \overline{X}^{1''}, \overline{X}^{2''}] \in \mathbb{R}^{[3\times(w\times B)\times n\times(ws,ws)\times(C/n)]}$.

Additionally, we assume that a larger cosine distance between the data of different modalities indicates a higher likelihood of complementary features. Therefore, we first compute the cosine similarity matrix between them using the following formulas:

$$\mathcal{S}^0 = \text{CosSim}\left[\overline{X}^{0'}, (\overline{X}^{1'} + \overline{X}^{2'})\right] = \frac{\left(\overline{X}^{1'} + \overline{X}^{2'}\right)\left(\overline{X}^{0'}\right)^{\text{T}}}{\left\|\left(\overline{X}^{1'} + \overline{X}^{2'}\right)\right\|\left\|\overline{X}^{0'}\right\|}, \tag{7}$$

$$\mathcal{S}^1 = \text{CosSim}\left[\overline{X}^{1'}, (\overline{X}^{0'} + \overline{X}^{2'})\right] = \frac{\left(\overline{X}^{0'} + \overline{X}^{2'}\right)\left(\overline{X}^{1'}\right)^{\text{T}}}{\left\|(\overline{X}^{0'} + \overline{X}^{2'})\right\|\left\|\overline{X}^{1'}\right\|}, \tag{8}$$

$$\mathcal{S}^2 = \text{CosSim}\left[\overline{X}^{2'}, (\overline{X}^{0'} + \overline{X}^{1'})\right] = \frac{\left(\overline{X}^{0'} + \overline{X}^{1'}\right)\left(\overline{X}^{2'}\right)^{\text{T}}}{\left\|(\overline{X}^{0'} + \overline{X}^{1'})\right\|\left\|\overline{X}^{2'}\right\|}. \tag{9}$$

Following the Transformer structure, to identify the differences between $x^0$ and $[x^1, x^2]$, we generate $Q$ and $V$ from the linear transformation of $x^0$, while $K$ is derived from the sum of the linear transformations of $x^1$ and $x^2$. This approach is similarly extended to other cases. After obtaining the cosine similarity matrix, we calculate the cosine distance and assign it to their respective modalities. The computational procedure is delineated as follows:

$$\widetilde{X^i} = \text{Softmax}(1 - \text{Softmax}(\mathcal{S}^i))\overline{X}^{i''}, i \in [0, 1, 2]. \tag{10}$$

The operation $\text{Softmax}(1-\text{Softmax}(\cdot))$ results in a normalized cosine distance matrix that transforms similarity into dissimilarity. This matrix is then used in a dot product with the original modality data to establish dissimilarity relationships.

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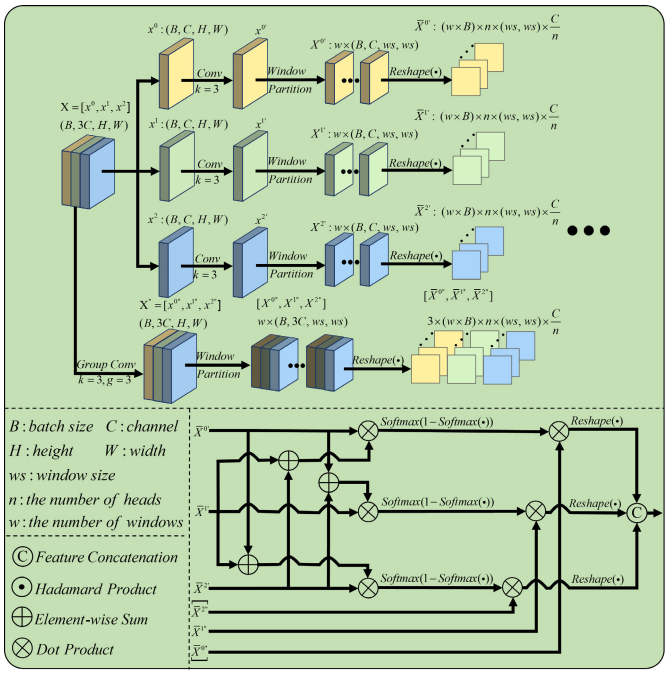



Fig. 4. Illustration of the proposed MLF module. The combined multimodal data is divided into its own modal data and then divided into several small windows using splitting to reduce the computational amount of subsequent operations. Then, the difference information is obtained by solving the cosine distance between different modal data. Subsequently, the difference information undergoes a dot product with the original modality data to assign dissimilarity relationships.

Through these operations, we effectively capture the differences between modalities.

*4) LLF Module:* As the number of layers increases, a semantic gap exists between the feature maps of different layers. To effectively reduce this gap, we propose the LLF module, as depicted in Fig. 2. This module fuses features from both spatial and channel aspects. Given an intermediate feature map $X_i \in \mathbb{R}^{B\times C_i\times H_i\times W_i}$ for $i \in (1, 2, 3, 4, 5)$, the LLF module successively generates a 2-D spatial attention map $Att^S$ and a 1-D channel attention map $Att^C$. The process can be summarized as follows:

$$\widehat{X_i} = \text{Concat}[\text{AvgP}_S(X_i), \text{MaxP}_S(X_i)], \tag{11}$$

$$\text{Att}_i^S = \sigma(\text{Shared_Conv2D}[\widehat{X_1}, \ldots, \widehat{X_i}, \ldots, \widehat{X_5}]), \tag{12}$$

$$X_i' = X_i + X_i \odot Att_i^S, \tag{13}$$

$$X_{\text{Aug}} = \text{Concat}[\text{AvgP}_C(X_1'), \ldots, \text{AvgP}_C(X_5')], \tag{14}$$

$$\text{Att}_i^C = \sigma(\text{Conv1D}(\text{FC}_i(X_{\text{Aug}}))), \tag{15}$$

$$X_i'' = X_i' + X_i' \odot \text{Att}_i^C \tag{16}$$

where $\text{AvgP}_S$ and $\text{AvgP}_C$ denote average pooling layers that retain only spatial and channel dimensions, respectively. $\text{MaxP}_S$ denotes a max-pooling layer that retains only the spatial dimension. Shared_Conv2D represents a 2-D dilated convolution, and $\text{FC}_i$ represent FC layer at the $i$th layer.

In terms of spatial processing, the module integrates multilayer and multiscale information along the spatial axis to create attention maps for each layer. Initially, the feature maps from each layer undergo separate average and max-pooling to retain only the spatial dimensions. A cyclic input approach is then employed, utilizing shared dilated convolution Shared_Conv2D to integrate spatial information

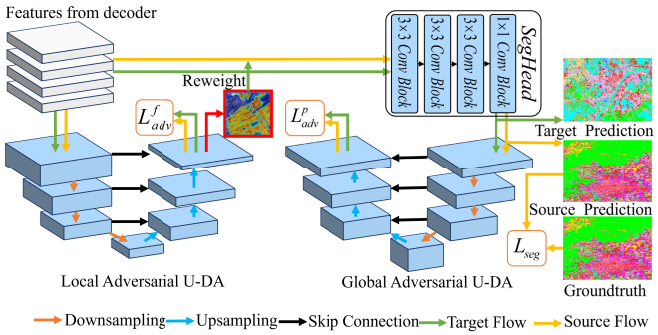



Fig. 5. Illustration of the proposed LGADA module and the SegHead module. The feature data (source and target domains) received from the decoder at the end of stage one, through local adversarial U-DA and global adversarial U-DA, narrow the gap between the source-domain and target-domain data by local alignment to guide global alignment. The final prediction output is obtained by the SegHead module.

across different stages with parameters $d = 3$ and $k = 7$. For channel processing, the module integrates multilayer information along the channel axis to generate attention maps for each layer. The feature map of each layer first undergoes average pooling to preserve only the channel dimension. The fusion process is then divided into local information fusion (using 1-D convolution operations) and global information fusion (utilizing different FC layers for each layer), aiming to produce more informative attention feature maps.

### C. Stage Two: Adversarial DA Network of the Fused Feature

The second stage aims to bridge the semantic gap between the source and target domains through a two-stage adversarial learning approach, culminating in the training of a SegHead module to predict segmentation results for the target images.

*1) LGADA Module:* Traditional adversarial networks often rely on a global adversarial training process, which can lead to negative information transfer and potentially degrade model performance in well-aligned regions [34]. To address this issue, we designed the LGADA module, which is illustrated in Fig. 5. This module consists of two DA modules: the local adversarial U-shaped DA (U-DA) module $F$ and the global adversarial U-DA module $P$. Initially, local adversarial training is performed to create a local alignment attention map, enabling the assignment of different weights to regions based on their local alignment. Following this, global adversarial training is conducted to achieve alignment of information between the source and target domains.

Specifically, the source-domain data $x_S$ and target-domain data $x_T$ are processed through the network of stage one $E$ to generate new source-domain data $E_S = E(x_S)$ and new target-domain data $E_T = E(x_T)$. These transformed data are then fed into $F$ for local adversarial learning. Through gradual optimization of the local adversarial loss $L_{\text{adv}}^f$, it aligns locally between $E_S$ and $E_T$ while assigning confidence scores to $E_T$ to generate an attention map $\alpha = |F(E_T)|$. Simultaneously, the source-domain data $x_S$ processed by $E$ and the target-domain data $x_T$ processed by $E$ and weighted by $\alpha$ are input into the SegHead module for prediction. The outputs obtained are $S_S = \text{SegHead}(E_S)$ and $S_T = \text{SegHead}(E_T + \alpha \times E_T)$, respectively. On the one hand, the output $S_S$ is then used to calculate the

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segmentation loss $L_{\text{seg}}$ with the source-domain data labels $y_S$. On the other hand, $S_S$ and $S_T$ are fed into $P$ for adversarial learning. By gradually optimizing the global adversarial loss $L^p_{\text{adv}}$, global alignment is performed between $S_S$ and $S_T$.

*2) SegHead Module:* As shown in Fig. 5, the SegHead module processes the feature maps $E_S$ and $E_T$ from the source and target domains following DA. For the source-domain data $E_S$, it produces a corresponding feature map $S_S$ that aligns with the source-domain labels and computes the associated loss. In contrast, for the target-domain data $E_T$, it reconstructs and generates the final predicted image $S_T$. This process can be mathematically expressed as

$$S_S = \text{SegHead}(E_S), \tag{17}$$

$$S_T = \text{SegHead}(E_T + \alpha \times E_T) \tag{18}$$

where the function SegHead represents the SegHead module, which includes convolutional layers, batch normalization (BN), and ReLU activation functions.

### D. Loss Function

The overall loss function is composed of three key components. The first term pertains to the segmentation loss, which is derived from the discrepancy between the predictions and the actual labels for the source-domain data. This includes both multiclass cross-entropy loss and Dice loss. The second term involves a local adversarial loss between the source and target domains, while the third term encompasses a global adversarial loss between them. The complete formula is represented as

$$L_{\text{all}} = L_{\text{seg}} + \lambda L^f_{\text{adv}} + \mu L^p_{\text{adv}} \tag{19}$$

where $\lambda$ and $\mu$ are hyperparameters that help balance the significance of different loss components during training, both set to 0.5 based on empirical observations. The segmentation loss term is defined as

$$L_{\text{seg}} = \mathbb{E}_{\hat{y_S}, y_S}[L_{\text{MCE}}(\hat{y_S}, y_S) + L_{\text{Dice}}(\hat{y_S}, y_S)]. \tag{20}$$

The average pixel loss $L_{\text{MCE}}$ is calculated as follows:

$$L_{\text{MCE}} = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_S^{i,c} \log(\hat{y_S}^{i,c}). \tag{21}$$

The Dice loss $L_{\text{Dice}}$ is formulated to address the issue of imbalanced training samples

$$L_{\text{Dice}} = 1 - \frac{2}{N \times C}\sum_{i=1}^{N}\sum_{c=1}^{C}\frac{\hat{y_S}^{i,c} \times y_S^{i,c}}{\hat{y_S}^{i,c} + y_S^{i,c}} \tag{22}$$

where $N$ denotes the number of classes, $C$ represents the number of samples, $y_S$ indicates the label of the source-domain data, and $\hat{y_S} = \text{SegHead}(E(x_S))$ represents the predicted output of the network after the function of $E$ and SegHead modules.

The second and third terms correspond to local and global adversarial losses, respectively. For the local adversarial U-DA module, $E_S$ and $E_T$ obtained by $x_S$ and $x_T$ through the action of $E$ are input into $F$ for adversarial learning. By performing continuous alternating optimization between $F$ and $E$, the loss $L^f_{\text{adv}}$ is minimized, represented as

$$\begin{aligned} L^f_{\text{adv}}(F) &= \mathbb{E}_{x_S}[(F(E_S) - 0)^2] + \mathbb{E}_{x_T}[(F(E_T) - 1)^2], \\ L^f_{\text{adv}}(E) &= \mathbb{E}_{x_T}[(F(E(x_T)) - 0)^2]. \end{aligned} \tag{23}$$

For the global adversarial U-DA module, the predictions $S_T$ and $S_S$ engage in adversarial learning within $P$, where the loss $L^p_{\text{adv}}$ is minimized through continuous alternating optimization between $P$ and SegHead$(E(\cdot))$, represented as

$$\begin{aligned} L^p_{\text{adv}}(P) &= \mathbb{E}_{x_S}[(P(S_S) - 0)^2] + \mathbb{E}_{x_T}[(P(S_T) - 1)^2], \\ L^p_{\text{adv}}(E, \text{SegHead}) &= \mathbb{E}_{x_T}[(P(\text{SegHead}(E(x_T)) - 0)^2]. \end{aligned} \tag{24}$$

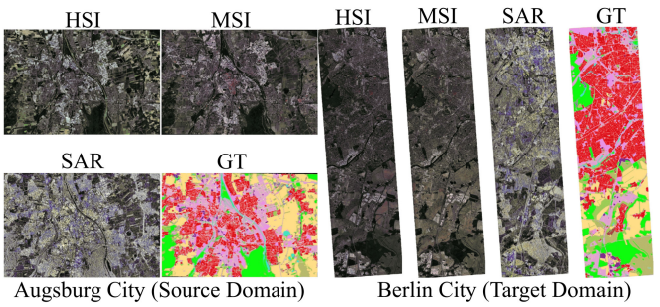



Fig. 6. C2Seg-AB data visualization of semantic segmentation research scenarios in Augsburg and Berlin based on multimodal RS data.

## IV. Experiments and Analysis

In this section, we present the design and results of our experiments. We begin by detailing the benchmark datasets and the experimental setup. Then, we conducted experiments on the datasets and compared the results with those of advanced methods.

### A. Dataset Descriptions

*1) C2Seg-AB Dataset:* The dataset provides multimodal RSI data and labeled semantic classes for the cities of Augsburg and Berlin in Germany [32]. It consists of EnMAP hyperspectral images (HSIs), Sentinel-2 multispectral images (MSIs), and Sentinel-1 SAR images. The HSI consists of 242 spectral bands, while the MSI data encompasses four spectral bands. The SAR data features dual-polarized images with HH and HV channels. All three data types are aligned pixel by pixel, with dimensions of $1360 \times 886$ pixels for Augsburg and $811 \times 2465$ pixels for Berlin, ensuring a consistent spatial resolution.

*2) C2Seg-BW Dataset:* The dataset provides multimodal RSI data and labeled semantic classes for the cities of Beijing and Wuhan in China [32]. It includes Gaofen-5 HSI data, Gaofen-6 MSI data, and Gaofen-3 SAR data. The HSI data covers 116 spectral channels, the MSI data consists of four spectral bands and the SAR data features dual-polarized imagery with HH and HV channels. The dimensions for these three data types are $8706 \times 13\,474$ pixels for Beijing and $8670 \times 6225$ pixels for Wuhan, maintaining a consistent spatial resolution.

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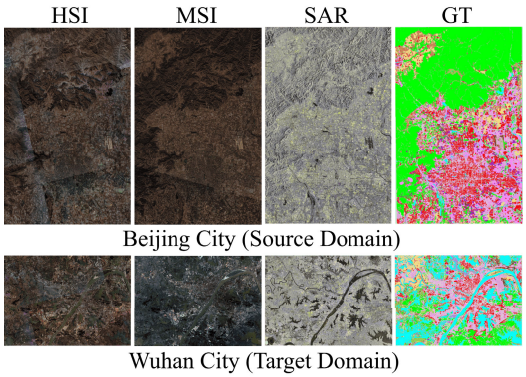



Fig. 7. C2Seg-BW data visualization of semantic segmentation research scenarios in Beijing and Wuhan based on multimodal RS data.

TABLE I
KEY FEATURES OF THE C2SEG-AB AND C2SEG-BW DATASETS

| Name | C2Seg-AB | C2Seg-BW |
|---|---|---|
| Spatial size | Augburg: 1360× 886<br>Berlin: 811×2465 | Beijing:8706×13474<br>Wuhan: 8670×6225 |
| Spatial resolution | 10m | 10m |
| Number of bands | HSI: 242 bands<br>MSI: 4 bands<br>SAR: 2 bands | HSI: 116 bands<br>MSI: 4 bands<br>SAR: 2 bands |
| Size of patches | 128× 128 | 256× 256 |
| Number of classes | 13 | 13 |

*3) Berlin–Augsburg:* To verify the robustness and superiority of the model, we utilized the multimodal RS data of Augsburg as the test set and the Berlin scene as the training set, named the Berlin–Augsburg dataset, and added it to the comparison experiment.

Thirteen distinct land-use and land-cover categories were classified using these datasets, with corresponding ground-truth (GT) data provided for validation. The visualization of C2Seg-AB and C2Seg-BW datasets is presented in Figs. 6 and 7, respectively. The main characteristics of both datasets are summarized in Table I. As the Berlin–Augsburg dataset shares an identical composition with the C2Seg-AB dataset, no additional description is provided.

### B. Implementation Details

*1) Experimental Settings:* The MC-Seg model was developed using the PyTorch framework. The experiments were conducted on a system equipped with an Intel Xeon Gold 6250 CPU, 128 GB of RAM, and a 32 GB NVIDIA Quadro GV100 GPU. For the C2Seg-AB dataset, we employed a sliding window technique to divide the original images into patches of 128× 128 pixels. This process yielded 273 source-domain images and 140 target-domain images. The training process lasted for 6000 epochs. For the Berlin–Augsburg dataset, a similar dividing process was adopted. This process produces 140 source-domain images and 273 target-domain images. The training process lasted 8000×. For the C2Seg-BW dataset, a similar sliding window approach was adopted, but with a patch size of 256 × 256, resulting in 7140 source-domain images and 850 target-domain images. Training proceeded for 10 000 epochs. A batch size of 8 and a learning rate of 0.0001 were maintained in all datasets. The Adam optimizer [35] was used for training. Given the high dimensionality and large size of the HSI data, we applied principal component analysis to preprocess the HSI data, reducing its dimensionality to ten channels.

To evaluate segmentation performance, we utilized overall accuracy (OA), mean intersection with union (mIoU), and mean $F1$ score (mF1) as evaluation metrics [32]. Specifically, OA reflects the global precision across all samples, mIoU measures the accuracy for each class, and mF1 evaluates the balanced performance of precision and recall.

The architecture of the MC-Seg model is described in detail, layer by layer. The first stage consists of five sequential layers incorporating CFE modules, FLF modules, and 2 × 2 downsampling operations. This is followed by five parallel MLF modules and an LLF module. The next phase includes five layers of consecutive 2 × 2 upsampling operations and CFE modules. The second stage features an LGADA module and a SegHead module. Table II presents the layered configuration of the MC-Seg model.

*2) Parameter Tuning:* The learning rate, batch size, and epochs are critical hyperparameters in deep learning frameworks. In specific, the learning rate significantly influences the optimization process; a minimal learning rate may result in slow convergence, while an overly large learning rate can cause gradient oscillations around the minimum, potentially preventing network convergence. The batch size directly impacts training speed, model generalization, and stability. A large batch size may result in memory issues, whereas a small batch size can introduce training instability, gradient fluctuations, and underfitting risks. The number of epochs is crucial for model performance and training outcomes, as an appropriate selection can balance accuracy and generalization while mitigating overfitting or underfitting.

To determine the optimal hyperparameters, we conducted experiments to identify suitable learning rates, batch sizes, and epochs. Fig. 8 illustrates the segmentation performance of the framework across the three datasets under varying learning rates {0.001, 0.0001, 0.00001}, batch sizes {2, 4, 8}, and epochs. The $x$-axis represents the batch sizes, the $y$-axis denotes the learning rates, the $z$-axis corresponds to the number of epochs, and the bubble diameter reflects the mIoU values. The optimal learning rate for all three datasets was found to be 0.0001, with a batch size of 8 (as selecting 16 or larger caused memory issues during training). The appropriate epochs for the respective datasets were determined to be 6000, 10 000, and 8000.

### C. Quantitative Evaluation of Comparative Experiments

To validate the advancement and effectiveness of the proposed MC-Seg, we conducted both qualitative and quantitative comparisons with classical and advanced segmentation models. The classical segmentation models include UNet [36] and DeepLab [37]. The advanced models are divided into

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TABLE II
NETWORK CONFIGURATION OF THE MC-SEG

| | | | HSI | MSI | SAR | Input channel | Input size ratio | Output channel | Output size ratio |
|---|---|---|---|---|---|---|---|---|---|
| Stage One | Feature Encoding, Fusion and downsampling | | CFE module | CFE module | CFE module | [10,4,2] | 1 | [16,16,16] | 1 |
| | | | FLF module | FLF module | FLF module | [16,16,16] | 1 | [16,16,16] | 1 |
| | | | 2×downsampling | 2×downsampling | 2×downsampling | [16,16,16] | 1 | [16,16,16] | 1/2 |
| | | | CFE module | CFE module | CFE module | [16,16,16] | 1/2 | [32,32,32] | 1/2 |
| | | | FLF module | FLF module | FLF module | [32,32,32] | 1/2 | [32,32,32] | 1/2 |
| | | | 2×downsampling | 2×downsampling | 2×downsampling | [32,32,32] | 1/2 | [32,32,32] | 1/4 |
| | | | CFE module | CFE module | CFE module | [32,32,32] | 1/4 | [64,64,64] | 1/2 |
| | | | FLF module | FLF module | FLF module | [64,64,64] | 1/4 | [64,64,64] | 1/4 |
| | | | 2×downsampling | 2×downsampling | 2×downsampling | [64,64,64] | 1/4 | [64,64,64] | 1/8 |
| | | | CFE module | CFE module | CFE module | [64,64,64] | 1/8 | [128,128,128] | 1/8 |
| | | | FLF module | FLF module | FLF module | [128,128,128] | 1/8 | [128,128,128] | 1/8 |
| | | | 2×downsampling | 2×downsampling | 2×downsampling | [128,128,128] | 1/8 | [128,128,128] | 1/16 |
| | | | CFE module | CFE module | CFE module | [128,128,128] | 1/16 | [256,256,256] | 1/16 |
| | | | FLF module | FLF module | FLF module | [256,256,256] | 1/16 | [256,256,256] | 1/16 |
| | Modality Fusion | | | MLF module | | 16×3 | 1 | 16×3 | 1 |
| | | | | MLF module | | 32 × 3 | 1/2 | 32 ×3 | 1/2 |
| | | | | MLF module | | 64 × 3 | 1/4 | 64 ×3 | 1/4 |
| | | | | MLF module | | 128 × 3 | 1/8 | 128×3 | 1/8 |
| | | | | MLF module | | 256 × 3 | 1/16 | 256×3 | 1/16 |
| | Layer Fusion | | | LLF module | | 16 × 3<br>32 × 3<br>64 × 3<br>128 × 3<br>256 × 3 | 1<br>1/2<br>1/4<br>1/8<br>1/16 | 16 × 3<br>32 × 3<br>64 × 3<br>128 × 3<br>256 × 3 | 1<br>1/2<br>1/4<br>1/8<br>1/16 |
| | Feature Decoding and up-sampling | | CFE module + 2×upsampling | | | 256×3 | 1/16 | 256×3 | 1/8 |
| | | | CFE module + 2×upsampling | | | 384×3 | 1/8 | 384×3 | 1/4 |
| | | | CFE module + 2×upsampling | | | 448×3 | 1/4 | 448×3 | 1/2 |
| | | | CFE module + 2×upsampling | | | 480×3 | 1/2 | 480×3 | 1 |
| | | | CFE module | | | 496×3 | 1 | 496×3 | 1 |
| Stage Two | Adversarial domain adaptation | Local Adversarial U-DA | 3×3Conv+BN+ReLU+2×downsampling | | | 1488 | 1 | 256 | 1/2 |
| | | | 3×3Conv+BN+ReLU+2×downsampling | | | 256 | 1/2 | 128 | 1/4 |
| | | | 3×3Conv+BN+ReLU+2×downsampling | | | 128 | 1/4 | 64 | 1/8 |
| | | | 3×3Conv+BN+ReLU | | | 64 | 1/8 | 32 | 1/8 |
| | | | 2×upsampling+3×3Conv+BN+ReLU | | | 32+64 | 1/8 | 64 | 1/4 |
| | | | 2×upsampling+3×3Conv+BN+ReLU | | | 64+128 | 1/4 | 32 | 1/2 |
| | | | 2×upsampling+3×3Conv+BN+ReLU | | | 32+256 | 1/2 | 16 | 1 |
| | | | 3×3Conv+BN+ReLU | | | 16 | 1 | 1 | 1 |
| | | Global Adversarial U-DA | 3×3Conv+BN+ReLU+2×downsampling | | | Number of Class | 1 | 32 | 1/2 |
| | | | 3×3Conv+BN+ReLU+2×downsampling | | | 32 | 1/2 | 64 | 1/4 |
| | | | 3×3Conv+BN+ReLU+2×downsampling | | | 64 | 1/4 | 128 | 1/8 |
| | | | 3×3Conv+BN+ReLU | | | 128 | 1/8 | 256 | 1/8 |
| | | | 2×upsampling+3×3Conv+BN+ReLU | | | 256+128 | 1/8 | 128 | 1/4 |
| | | | 2×upsampling+3×3Conv+BN+ReLU | | | 128+64 | 1/4 | 64 | 1/2 |
| | | | 2×upsampling+3×3Conv+BN+ReLU | | | 64+32 | 1/2 | 32 | 1 |
| | | | 3×3Conv+BN+ReLU | | | 32 | 1 | 1 | 1 |
| | Segmentation Head | SegHead | 3×3Conv+BN+ReLU | | | 1488 | 1 | 512 | 1 |
| | | | 3×3Conv+BN+ReLU | | | 512 | 1 | 128 | 1 |
| | | | 3×3Conv+BN+ReLU | | | 128 | 1 | 32 | 1 |
| | | | 1×1Conv+BN+ReLU | | | 32 | 1 | Number of Class | 1 |

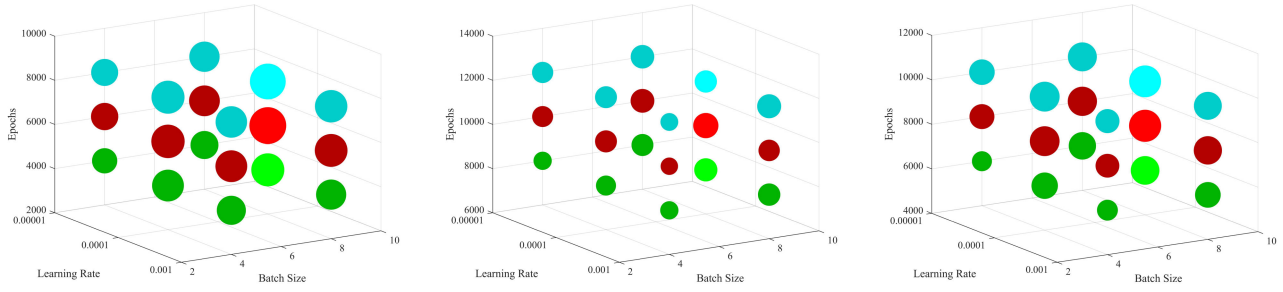



Fig. 8. Comparison of model performance across varying learning rates, batch sizes, and epochs. (a) C2Seg-AB. (b) C2Seg-BW. (c) Berlin-Augsburg.

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TABLE III

QUANTITATIVE PERFORMANCE COMPARISON OF SEGMENTATION NETWORKS IN TERMS OF OA, MIOU, MF1, AND IOU FOR EACH CLASS ON C2SEG-AB DATASET. THE SYMBOL "／" DENOTES NO PIXELS CORRECTLY IDENTIFIED. THE BEST RESULT IS MARKED IN BOLD

| Class Name | Classical Model | | Advanced Model without DA | | Advanced Model with DA | | | | Our Model |
|---|---|---|---|---|---|---|---|---|---|
| | UNet | DeepLab | DualHR | SegFormer | AdaptSeg | DSAN | MBATA-GAN | HighDAN | MC-Seg |
| *Surface Water* | ／ | ／ | 0.1759 | 0.3387 | 0.1289 | 0.1126 | 0.3453 | 0.3662 | **0.3909** |
| *Street Network* | 0.0360 | 0.0023 | 0.0104 | 0.0772 | 0.0910 | 0.0209 | 0.0162 | 0.0863 | **0.1940** |
| *Urban Fabric* | 0.4843 | 0.5080 | 0.5023 | 0.4925 | 0.4714 | 0.5301 | 0.5753 | 0.5916 | **0.5969** |
| *Industrial, Commercial and Transport* | 0.3097 | 0.3274 | 0.3676 | 0.3236 | 0.2998 | 0.2060 | 0.4298 | 0.4481 | **0.4713** |
| *Mine, Dump, and Construction Sites* | 0.1003 | 0.0001 | ／ | 0.1218 | 0.0193 | 0.0184 | **0.1332** | 0.0787 | 0.0833 |
| *Artificial Vegetated Areas* | 0.1548 | 0.1872 | 0.1344 | 0.2159 | 0.0690 | 0.1176 | **0.2166** | 0.2129 | 0.2016 |
| *Arable Land* | 0.3856 | 0.3831 | 0.4399 | 0.4693 | 0.3741 | 0.3969 | 0.4567 | 0.3833 | **0.4724** |
| *Permanent Crops* | ／ | ／ | ／ | ／ | ／ | ／ | ／ | 0.0042 | **0.0077** |
| *Pastures* | 0.0507 | ／ | 0.0530 | 0.0358 | 0.0322 | 0.1789 | 0.0324 | **0.2589** | 0.2448 |
| *Forests* | 0.1057 | 0.3600 | 0.3764 | 0.5021 | 0.4548 | 0.4410 | 0.5927 | 0.5671 | **0.6525** |
| *Shrub* | 0.0035 | ／ | ／ | 0.0011 | 0.0031 | 0.0014 | 0.0021 | **0.0990** | 0.0827 |
| *Open Spaces with no Vegetation* | ／ | ／ | ／ | ／ | 0.0006 | ／ | ／ | ／ | **0.0025** |
| *Inland Wetlands* | ／ | ／ | ／ | ／ | ／ | ／ | ／ | 0.0002 | **0.0005** |
| $OA$ | 0.4226 | 0.4344 | 0.4726 | 0.5114 | 0.4682 | 0.4985 | 0.5561 | 0.5689 | **0.6003** |
| $mIoU$ | 0.1254 | 0.1360 | 0.1585 | 0.1983 | 0.1496 | 0.1557 | 0.2154 | 0.2382 | **0.2616** |
| $mF1$ | 0.1920 | 0.1977 | 0.2324 | 0.2884 | 0.2234 | 0.2316 | 0.3049 | 0.3409 | **0.3818** |
| *Time Consumed* (ms) | **41** | 145 | 108 | 852 | 127 | 174 | 678 | 265 | 354 |

TABLE IV

QUANTITATIVE PERFORMANCE COMPARISON OF SEGMENTATION NETWORKS IN TERMS OF OA, MIOU, MF1, AND IOU FOR EACH CLASS ON THE C2SEG-BW DATASET. THE SYMBOL "／" DENOTES NO PIXELS CORRECTLY IDENTIFIED. THE BEST RESULT IS MARKED IN BOLD

| Class Name | Classical Model | | Advanced Model without DA | | Advanced Model with DA | | | | Our Model |
|---|---|---|---|---|---|---|---|---|---|
| | UNet | DeepLab | DualHR | SegFormer | AdaptSeg | DSAN | MBATA-GAN | HighDAN | MC-Seg |
| *Surface Water* | ／ | 0.2257 | 0.3790 | 0.4743 | 0.2789 | 0.0423 | 0.3992 | 0.5948 | **0.6233** |
| *Street Network* | 0.0230 | 0.0036 | 0.0040 | 0.0106 | 0.0247 | 0.0148 | 0.0185 | 0.0042 | **0.0944** |
| *Urban Fabric* | 0.1462 | 0.1175 | 0.1388 | 0.2761 | 0.1392 | 0.2316 | 0.2791 | **0.2832** | 0.2035 |
| *Industrial, Commercial and Transport* | 0.1673 | 0.1094 | 0.2526 | 0.2422 | 0.1821 | 0.2634 | 0.2473 | 0.2323 | **0.2952** |
| *Mine, Dump, and Construction Sites* | **0.0169** | 0.0111 | 0.0058 | 0.0077 | 0.0107 | 0.0020 | 0.0024 | 0.0047 | 0.0060 |
| *Artificial Vegetated Areas* | 0.0255 | 0.0229 | 0.0027 | 0.0288 | 0.0499 | 0.0405 | 0.0324 | 0.0344 | **0.0648** |
| *Arable Land* | 0.0449 | 0.0032 | 0.0097 | 0.0603 | 0.0506 | 0.1057 | **0.1187** | 0.1086 | 0.0778 |
| *Permanent Crops* | 0.0001 | 0.0001 | 0.0008 | 0.0017 | 0.0016 | ／ | **0.0026** | 0.0006 | 0.0008 |
| *Pastures* | ／ | ／ | ／ | ／ | ／ | ／ | ／ | ／ | ／ |
| *Forests* | 0.1745 | 0.1948 | 0.0757 | 0.2501 | 0.01832 | 0.2105 | 0.2799 | 0.2563 | **0.3504** |
| *Shrub* | ／ | ／ | 0.0018 | 0.0005 | ／ | ／ | ／ | ／ | **0.0022** |
| *Open Spaces with no Vegetation* | ／ | ／ | **0.0023** | 0.0002 | ／ | ／ | ／ | ／ | 0.0005 |
| *Inland Wetlands* | ／ | ／ | **0.0004** | ／ | ／ | ／ | ／ | ／ | ／ |
| $OA$ | 0.1641 | 0.2117 | 0.3394 | 0.3529 | 0.2136 | 0.2041 | 0.3548 | 0.3782 | **0.4603** |
| $mIoU$ | 0.0461 | 0.0529 | 0.0672 | 0.1040 | 0.0708 | 0.0701 | 0.1062 | 0.1169 | **0.1322** |
| $mF1$ | 0.0795 | 0.0863 | 0.1085 | 0.1597 | 0.1201 | 0.1172 | 0.1643 | 0.1734 | **0.1957** |
| *Time Consumed* (ms) | **175** | 596 | 442 | 5536 | 524 | 833 | 3864 | 1534 | 2040 |

two categories: those without DA, such as SegFormer [38] and Dual-stream HR-Net (DualHR) [39], and those with DA, including AdaptSeg [40], DSAN [41], MBATA-GAN [42], and HighDAN [32]. To ensure fairness in all experiments, we utilized the optimal network configurations for all compared methods (as reported in the literature). Additionally, relevant parameters were slightly adjusted to suit the multimodal RS data segmentation experiments. All comparative models underwent the same preprocessing operations and training optimization strategies.

*1) Comparison of Different Segmentation Methods:* Tables III–V provide a quantitative comparison of the performance of the eight selected models and our proposed model on the C2Seg-AB, C2Seg-BW, and Berlin-Augsburg datasets, respectively. The time consumed per epoch for all methods is also provided.

The results indicate that advanced models (like Segformer, MBATA-GAN, and HighDAN) generally outperform traditional models (UNet and DeepLab) due to their superior architectural designs. Among the classic networks, the DeepLab shows slightly better overall performance than UNet. However, benefitting from its encoder–decoder structure, the UNet performs marginally better in small structural categories such as *Mine, Dump, and Construction Sites* and *Street Network*. In comparing advanced models, the presence of a DA strategy is not the sole determinant of model performance. Experiments on the C2Seg-AB and Berlin-Augsburg datasets reveal that advanced models with DA (AdaptSeg and DSAN) perform similarly to those without DA (DualHR), indirectly confirming the importance of multimodal information extraction and fusion. Nevertheless, in the more challenging C2Seg-BW dataset, advanced models with DA perform

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TABLE V

QUANTITATIVE PERFORMANCE COMPARISON OF SEGMENTATION NETWORKS IN TERMS OF OA, MIOU, MF1, AND IOU FOR EACH CLASS ON BERLIN-AUGSBURG DATASET. THE SYMBOL "／" DENOTES NO PIXELS CORRECTLY IDENTIFIED. THE BEST RESULT IS MARKED IN BOLD

| Class Name | Classical Model | | Advanced Model without DA | | Advanced Model with DA | | | | Our Model |
|---|---|---|---|---|---|---|---|---|---|
| | UNet | DeepLab | DualHR | SegFormer | AdaptSeg | DSAN | MBATA-GAN | HighDAN | MC-Seg |
| *Surface Water* | 0.2348 | 0.2079 | 0.1518 | 0.2333 | 0.0968 | 0.1136 | 0.1608 | 0.2621 | **0.3379** |
| *Street Network* | ／ | ／ | 0.0242 | 0.0348 | 0.0006 | 0.0004 | 0.1006 | 0.1182 | **0.1719** |
| *Urban Fabric* | 0.2075 | 0.2656 | 0.2925 | 0.2908 | 0.2998 | 0.3127 | 0.2729 | 0.3016 | **0.3315** |
| *Industrial, Commercial and Transport* | 0.2137 | 0.3085 | 0.2849 | 0.3195 | 0.3140 | 0.3219 | 0.2861 | 0.3424 | **0.3924** |
| *Mine, Dump, and Construction Sites* | 0.1043 | 0.0617 | 0.1179 | 0.1203 | 0.1812 | 0.1922 | 0.1042 | 0.2144 | **0.2549** |
| *Artificial Vegetated Areas* | 0.0807 | 0.1046 | 0.1221 | 0.1964 | 0.1768 | 0.1660 | 0.2881 | 0.2816 | **0.3037** |
| *Arable Land* | 0.0864 | 0.1081 | 0.4931 | 0.5428 | 0.2629 | 0.2354 | 0.3000 | 0.5682 | **0.6056** |
| *Permanent Crops* | ／ | ／ | ／ | ／ | 0.0004 | 0.0170 | 0.0134 | 0.0133 | **0.0212** |
| *Pastures* | ／ | ／ | 0.0745 | 0.1702 | 0.0912 | 0.1833 | 0.2085 | 0.2191 | **0.2662** |
| *Forests* | 0.4261 | 0.4354 | 0.3530 | 0.5192 | 0.3115 | 0.2508 | 0.4987 | 0.4883 | **0.5314** |
| *Shrub* | 0.0001 | ／ | 0.0005 | 0.0011 | 0.0312 | 0.0355 | 0.1860 | **0.1396** | 0.0990 |
| *Open Spaces with no Vegetation* | ／ | ／ | ／ | ／ | ／ | ／ | 0.0006 | 0.0017 | **0.0057** |
| *Inland Wetlands* | ／ | ／ | ／ | ／ | 0.0001 | 0.0001 | 0.0003 | 0.0008 | **0.0022** |
| $OA$ | 0.3614 | 0.3745 | 0.4379 | 0.4931 | 0.4022 | 0.4296 | 0.4867 | 0.5342 | **0.6117** |
| $mIoU$ | 0.1041 | 0.1148 | 0.1473 | 0.1868 | 0.1359 | 0.1407 | 0.1862 | 0.2270 | **0.2557** |
| $mF1$ | 0.1588 | 0.1624 | 0.2207 | 0.2679 | 0.2046 | 0.2133 | 0.2719 | 0.3287 | **0.3742** |
| *Time Consumed* (ms) | **39** | 152 | 104 | 845 | 125 | 182 | 667 | 270 | 341 |

slightly better overall than those without, emphasizing the significance of DA.

Among the advanced models without DA, the HR-Net architecture of the DualHR shows greater potential for extracting rich semantic information from multimodal RS data compared to CNN-based backbones. The achieved mIoU values were 15.85%, 6.72%, and 14.73% for the three datasets, respectively. It is worth noting that the Transformer-based SegFormer exhibits excellent performance on all three datasets, achieving improvements of 3.88% (1.35% and 5.52%), 3.98% (3.68% and 3.95%), and 5.6% (5.12% and 4.72%) in OA, mIoU, and mF1, respectively, compared to the DualHR on the C2Seg-AB (C2Seg-BW and Berlin-Augsburg) dataset.

Among the advanced models with DA, the AdaptSeg and DSAN utilize different DA schemes, with DSAN generally outperforming AdaptSeg on the C2Seg-AB and Berlin-Augsburg datasets, particularly in categories such as *Urban Fabric*, *Artificial Vegetated Areas*, and *Pastures*. However, the performance of these two models is similar in the C2Seg-BW dataset. In certain categories like *Surface Water* and *Street Network*, the mIoU values for DSAN are even slightly lower than those of AdaptSeg, which may be attributed to DSAN's limited ability to generalize on more challenging datasets. For the recently proposed MBATA-GAN with DA, thanks to its well-designed architecture, it achieves significant performance gains across all three datasets. Compared to DSAN on the C2Seg-AB (C2Seg-BW and Berlin-Augsburg) datasets, it demonstrates improvements of 5.76% (15.07% and 5.71%), 5.97% (3.61% and 4.55%), and 7.33% (4.71% and 5.86%) in OA, mIoU, and mF1, respectively.

It is important to recognize that the optimal results can be achieved through a combination of effective DA structures and fusion mechanisms. Among all the models compared, HighDAN stands out as the most competitive, being the latest multimodal cross-domain segmentation model. Despite not utilizing a Transformer structure, its design concepts for DF and DA allow it to maintain competitive and stable performance across all three datasets, securing the second-highest results in all metrics. When compared to the SegFormer on the C2Seg-AB dataset, the HighDAN shows improvements of approximately 5.75%, 3.99%, and 5.25% in OA, mIoU, and mF1, respectively.

Differences between the target and source domains arise due to variations in imaging conditions (e.g., lighting, atmosphere, and sensor parameters), leading to significant shifts in the statistical distribution of spectral reflectance and texture features. For instance, the spectral curves of the same object in the source domain and target domain may differ due to variations in water absorption, causing intramodal feature shifts. In the MC-Seg model, we address this by leveraging multimodal data and performing deep fusion across modalities, mitigating domain discrepancies through complementary information (e.g., the geometric information of SAR compensating for the limitations of HSIs) and overcoming the sensitivity of traditional models to distribution shifts. Similarly, images from two domains may be captured by different sensors, resulting in spatial resolution mismatches. The same object may appear as mixed pixels in low-resolution images but exhibit detailed structures in high-resolution images. To address this scale discrepancy, the MC-Seg model performs multilevel feature extraction on the input data from both domains, generating multiscale features. These features are then fused at three levels (feature-level, modal-level, and layer-level), ensuring full utilization of the input data and effectively alleviating spatial resolution mismatches. Finally, certain categories in the target domain may have significantly different frequencies compared to the source domain, leading to poor segmentation performance for low-frequency categories in the target domain. To mitigate this semantic discrepancy caused by category distribution shifts, the MC-Seg model employs a two-stage adversarial DA module. It guides global alignment through local alignment, effectively addressing discrepancies arising from category distribution shifts between the two domains.

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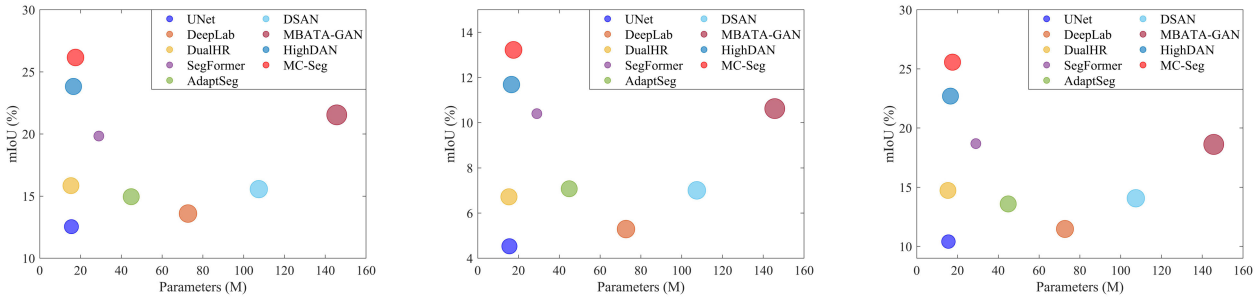



Fig. 9. Comparison of the networks on trainable parameters, computational cost, and mIoU. (a) C2Seg-AB. (b) C2Seg-BW. (c) Berlin-Augsburg.

As anticipated, the proposed MC-Seg model exhibits superior segmentation performance on the C2Seg-AB dataset, with enhancements of 3.14%, 2.34%, and 4.09% in OA, mIoU, and mF1 compared to HighDAN. Similar performance gains were observed on the C2Seg-BW and Berlin-Augsburg datasets, with OA increasing by 8.21% and 7.75%, mIoU improving by 1.53% and 2.87%, and mF1 enhancing by 2.23% and 4.55%. The experimental findings also indicate that MC-Seg achieves the highest mIoU scores in numerous categories across all three datasets, including *Surface Water*, *Street Network*, and *Forests*. However, it is worth mentioning that all three datasets are quite challenging for multimodal cross-domain segmentation, with C2Seg-BW being particularly difficult, resulting in many categories being poorly identified or yielding segmentation results close to 0%, such as *Permanent Crops*, *Open Spaces with no Vegetation*, and *Inland Wetlands*.

*2) Computational Complexity Analysis:* In addition to accuracy, it is important to take model complexity into account. Specifically, the number of parameters and the computational cost serve as effective measures of the spatial and temporal complexity of the network, respectively. Fig. 9(a)–(c) illustrates the mIoU, parameters counts, and computational costs for the various networks across the three datasets. The horizontal axis indicates the number of parameters, the vertical axis shows the mIoU, and the bubble diameter represents computational complexity (FLOPs).

Classic segmentation models like UNet have fewer parameters and lower computational costs, but they tend to perform relatively poorly. The AdaptSeg performs slightly better than the DeepLab, although both have more parameters than UNet while maintaining comparable computational costs and mIoU values. The DualHR model manages to enhance its performance while keeping parameter counts low, showcasing its significant advantage in extracting multimodal information due to its architecture. A notable feature of the SegFormer model is its ability to achieve relatively high segmentation accuracy at the lowest computational cost, despite having a greater number of parameters. Compared to the abovementioned models, the HighDAN significantly improves segmentation performance while still keeping the parameter count relatively low. Our proposed MC-Seg model, with a slight increase in parameter count compared to the HighDAN, achieves the best mIoU on all three datasets, with improvements of 2.34%, 1.53%, and 2.87%, respectively.

### *D. Visual Comparison on the C2Seg Datasets*

Figs. 10–12 depict the predictive segmentation maps generated by nine distinct segmentation models applied to the C2Seg-AB, C2Seg-BW, and Berlin-Augsburg datasets, respectively. The classic models (UNet and DeepLab) exhibit notable visual discrepancies compared to the GT. These models struggle to differentiate between categories such as *Street Network*, *Permanent Crops*, and *Pastures* and frequently misclassifying *Pastures* as *Arable Land* due to spectral similarities among these classes.

In contrast, the DualHR model shows a clear advantage over these classic models, demonstrating enhanced sensitivity to *Industrial, Commercial, and Transport* and more precise identification of *Arable Land*, although it tends to over-identify *Pastures* as *Arable Land*. The SegFormer, with its unique architecture, significantly surpasses the aforementioned models in recognizing *Surface Water*, *Industrial, Commercial, and Transport*, and *Forests*. Among the models utilizing DA, the AdaptSeg and DSAN show superior performance (except the SegFormer) in *Street Network* and *Forest*. Compared to the AdaptSeg and DSAN models, MBATA-GAN demonstrates significant performance enhancements across *Surface Water*, *Arable Land*, and *Forests* categories, attributed to its distinctive DA architecture. The latest HighDAN model marks a considerable advancement over previous models, providing clearer segmentation for *Surface Water*, *Urban Fabric*, *Industrial, Commercial, and Transport*, and *Forests*. Our proposed MC-Seg model exhibits superior performance, particularly in the categories of *Surface Water*, *Street Network*, and *Industrial, Commercial, and Transport*. However, it is important to note that our model faces challenges in distinguishing certain categories, such as *artificial, vegetated area*, *pasture*, *open, space, with, without, vegetation*, and *inland, wetland*.

Furthermore, we have included detailed segmentation results for the subregions of Fig. 10, as illustrated in Fig. 13. This localized comparison emphasizes the varying effectiveness of different models in capturing small target details. Among all the models compared, the classic models (UNet and DeepLab) primarily segment large objects, often losing critical information. This limitation highlights their challenges in handling cross-domain segmentation tasks, particularly in delineating large object boundaries and preserving small object details. While the SegFormer achieves commendable metrics among advanced models without DA, it struggles with capturing target

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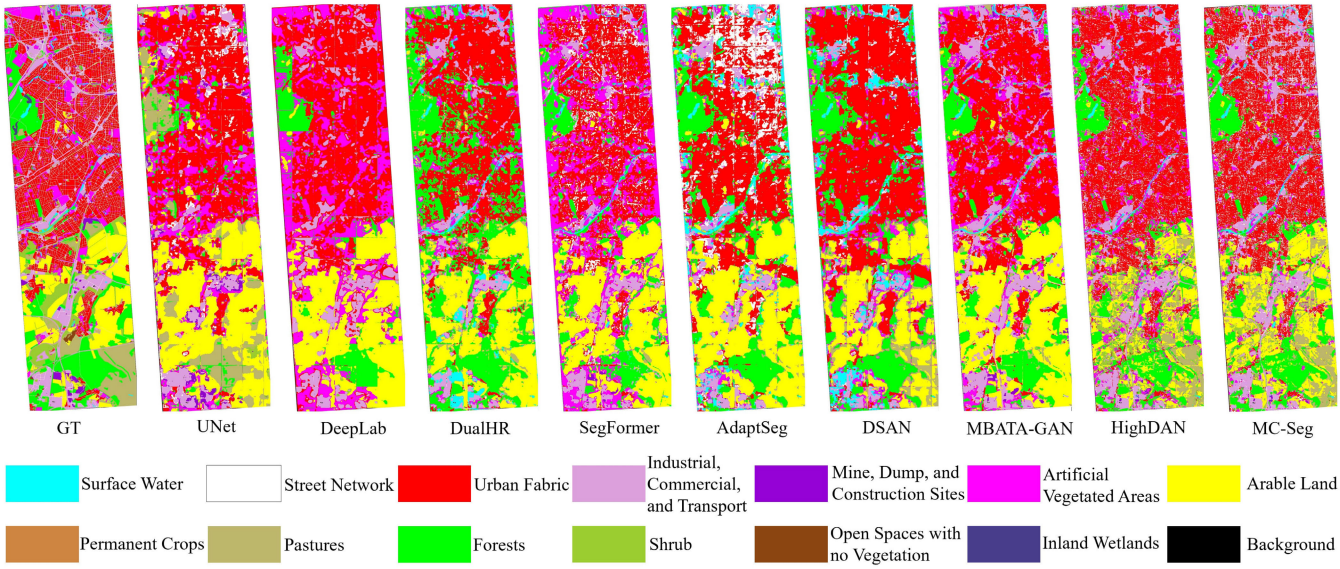



Fig. 10. Visualization of the semantic segmentation results obtained by successively using the UNet, DeepLab, DualHR, SegFormer, AdaptSeg, DSAN, MBATA-GAN, HighDAN, and our proposed MC-Seg on the C2Seg-AB (testing set: Berlin) dataset.

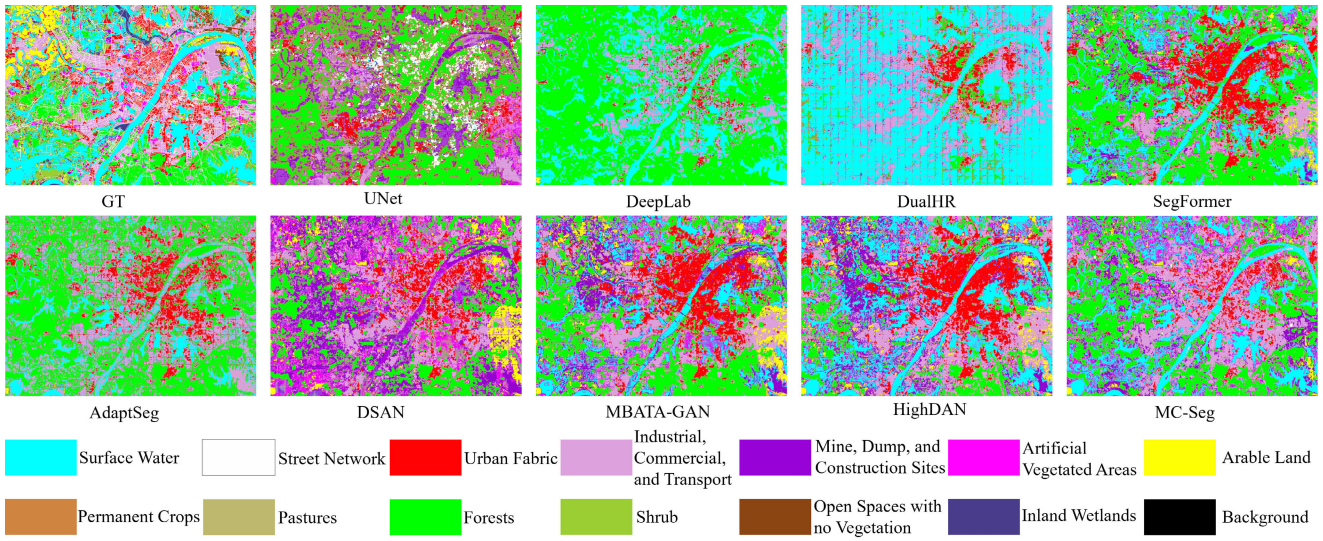



Fig. 11. Visualization of the semantic segmentation results obtained by successively using the UNet, DeepLab, DualHR, SegFormer, AdaptSeg, DSAN, MBATA-GAN, HighDAN, and our proposed MC-Seg on the C2Seg-BW (testing set: Wuhan) dataset.

details. Among the advanced models with DA, the AdaptSeg, DSAN, and MBATA-GAN show improvements in recognizing various semantic categories but still have limitations in boundary segmentation. The HighDAN outperforms these models, demonstrating competitiveness in capturing small and irregular objects. The visual comparison in Fig. 13 indicates that our MC-Seg model excels in targeting small objects and effectively captures fine details. By bridging the semantic gap between multimodal data and reducing the disparity between source and target domains through multimodal information fusion and adversarial DA, our model enhances segmentation precision. Nonetheless, it encounters difficulties in distinguishing categories with similar spectral characteristics, such as *Pastures* and *Arable Land*, leading to confusion between them. Additionally, it struggles to differentiate *Permanent Crops*, *Open Spaces with no Vegetation*, and *Inland Wetlands*. Overall, while our proposed model shows promising accuracy and robustness in multimodal cross-domain segmentation, there is still room for improvement in enhancing DA capabilities and multimodal information fusion.

### *E. Ablation Study*

To evaluate the significance of various key modules designed for cross-domain semantic segmentation of multimodal RSI data, we conducted ablation studies on the C2Seg-AB dataset. The proposed MC-Seg adopts the UNet architecture as its foundational framework while substituting conventional convolutional layers with CFE modules. Furthermore, it integrates three-level fusion modules (FLF, MLF, and LLF modules) and the LGADA module. All experimental models include a SegHead module by default. Since the design of the SegHead module is not particularly novel, it was excluded from the ablation analysis. Table VI provides detailed

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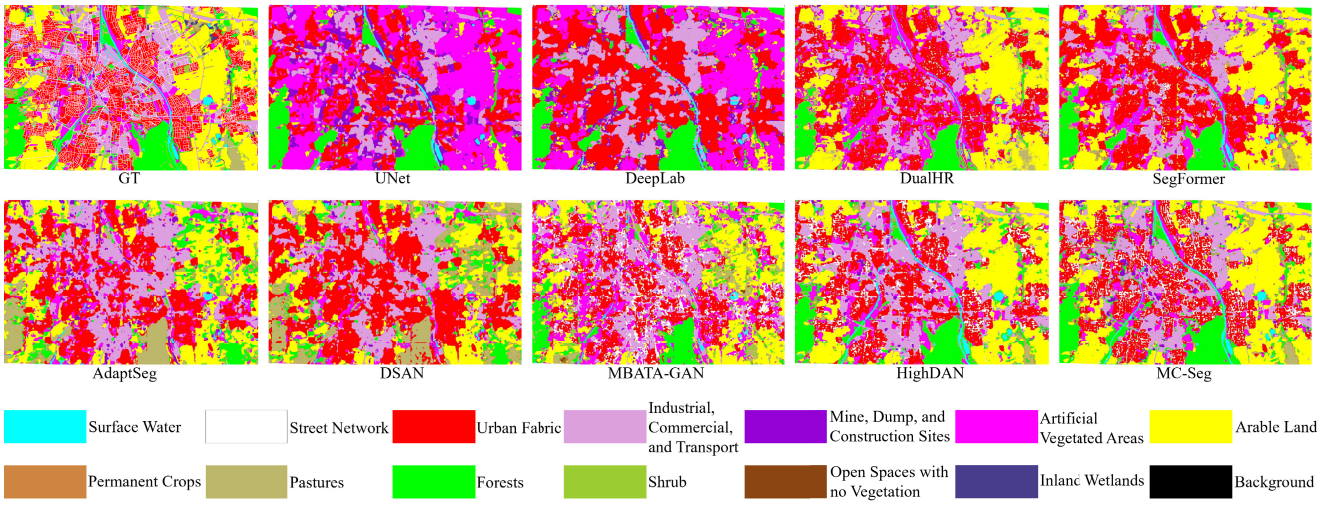



Fig. 12. Visualization of the semantic segmentation results obtained by successively using the UNet, DeepLab, DualHR, SegFormer, AdaptSeg, DSAN, MBATA-GAN, HighDAN, and our proposed MC-Seg on the Berlin-Augsburg (testing set: Augsburg) dataset.

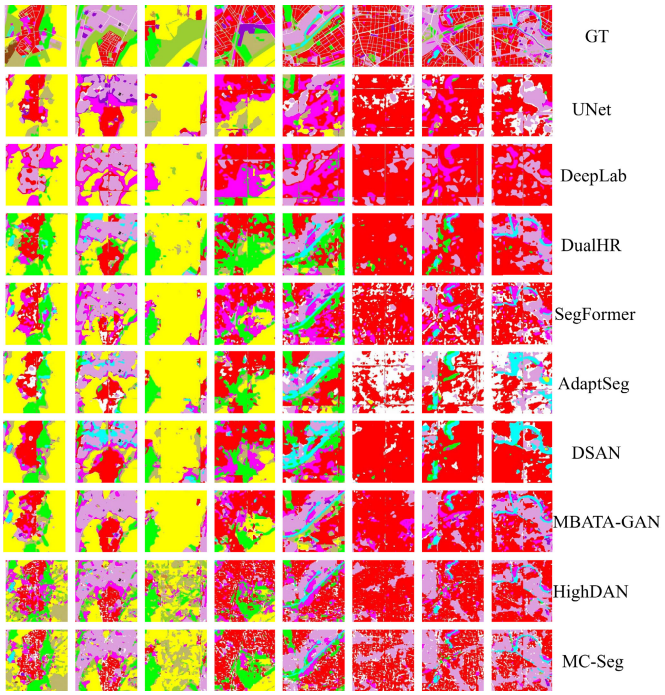



Fig. 13. Visualizing semantic segmentation results of subregions corresponding to Fig. 10 for all compared models.

TABLE VI

ABLATION ANALYSIS ON THE C2SEG-AB DATASETS. "CFE," "FLF," "MLF," AND "LLF" DENOTE THE CFE MODULE, FLF MODULE, MLF MODULE, AND LLF MODULE, RESPECTIVELY. "DA" DENOTE LGADA MODULE. THE BEST RESULTS ARE MARKED IN BOLD

| Modal | CFE | FLF | MLF | LLF | DA | $OA$ | $mIoU$ | $mF1$ |
|---|---|---|---|---|---|---|---|---|
| UNet-based | × | × | × | × | × | 0.4014 | 0.1113 | 0.1732 |
| UNet-based | √ | × | × | × | × | 0.4653 | 0.1604 | 0.2355 |
| UNet-based+Fusion | √ | √ | × | × | × | 0.5018 | 0.1824 | 0.2623 |
| UNet-based+Fusion | √ | × | √ | × | × | 0.5241 | 0.2006 | 0.2823 |
| UNet-based+Fusion | √ | × | × | √ | × | 0.4804 | 0.1783 | 0.2526 |
| UNet-based+Fusion | √ | √ | √ | √ | × | 0.5361 | 0.2215 | 0.3049 |
| UNet-based+DA | √ | × | × | × | √ | 0.5395 | 0.2221 | 0.3210 |
| UNet-based+Fusion+DA | √ | √ | √ | √ | √ | **0.6003** | **0.2616** | **0.3818** |

information on the performance gains (OA, mIoU and mF1) achieved by combining different components.

The UNet-based model, which serves as the baseline with a typical encoder–decoder structure, exhibits poor performance through standard convolutional modules for feature extraction. The replacement of these convolutional modules with our proposed CFE module significantly enhances performance across all metrics: OA increases by 6.39%, mIoU increases by 4.91%, and mF1 increases by 6.23%. This improvement is attributed to the CFE module's ability to capture both local and global features through its Transformer architecture, which facilitates long-range dependencies.

When the FLF module is combined with the CFE module during the encoding phase, it effectively fuses multimodal data while also extracting features from single-modality data, leading to notable improvements of 3.65% in OA, 2.2% in mIoU, and 2.68% in mF1. Additionally, the individual incorporation of the MLF and LLF modules into the UNet-based model yields significant performance gains compared to a model lacking these fusion modules, with OA increasing by 5.88% and 1.51%, respectively; mIoU improving by 4.02% and 1.79%, respectively; and mF1 rising by 4.68% and 1.71%, respectively. These enhancements demonstrate the superiority of the FLF, MLF, and LLF modules. It should be noted that integrating all three fusion modules into the network does not yield a significant improvement compared to adding each module individually. Specifically, when compared to the addition of the MLF module, the increases in OA, mIoU, and mF1 are only 1.2%, 2.09%, and 2.26%, respectively. This limited improvement is likely due to the lack of DA capabilities in the network, which creates performance bottlenecks in cross-domain tasks.

When conducting ablation studies with the DA module alone, we observe considerable improvements over the UNet-based model with the CFE module: OA, mIoU, and mF1 increased by 7.42%, 6.17%, and 8.55%, respectively. These

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results indicate that the LGADA module effectively bridges the gap between the source and target domains. Ultimately, the integration of all modules into the network results in our proposed model achieving optimal performance across all metrics, as shown in Table VI. This success is primarily due to the synergistic effects of the fusion strategies and DA techniques, which provide robust feature representations through the extraction and fusion of multimodal data, thereby facilitating effective DA processes.

## V. Conclusion

The combination of multimodal DF and DA technology can greatly diminish the gap between source and target domains and improve accuracy in multimodal cross-domain semantic segmentation tasks. To this end, we propose a novel multimodal cross-domain segmentation network of RSI called MC-Seg, which leverages multilevel deep cross-fusion and adversarial DA. Initially, the MC-Seg employs the CFE and FLF modules to effectively extract and fuse multimodal data, ensuring comprehensive utilization of features from various modalities. Subsequently, the MLF module identifies differences in multimodal data, while the LLF module integrates multiscale information between different layers. Finally, the LGADA module implements DA operations to reconcile the representational differences between the source and target domains. Extensive experiments conducted on the C2Seg dataset fully validate the effectiveness of the MC-Seg. In comparison analyses with other advanced models, the MC-Seg consistently outperforms all competitors on all metrics. In future work, we will focus on developing a more precise and lightweight multimodal cross-domain RSI segmentation model.

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